Optical Imaging and Holographic Object Extraction using Ray-Transfer Matrices

Derek Nester
McKelvey School of Engineering
Washington University in St. Louis
St. Louis, USA
derekn@wustl.edu

Naya Ford
McKelvey School of Engineering
Washington University in St. Louis
St. Louis, MO, USA
f.naya@wustl.edu

***Abstract*—This report details the simulation and analysis of optical imaging systems using linear algebra and ray-transfer matrices. The primary objective is to model the propagation of light rays through free space and lenses, and to apply these concepts to extract sharp images from a holographic light field dataset. The system is modeled using 4x4 ray-transfer matrices that linearize the behavior of light under the paraxial approximation. Part 1 validates the simulation by tracing rays through a finite lens system, demonstrating convergence and aperture clipping. Part 2 utilizes a computational autofocus algorithm based on contrast maximization to determine the optimal focal distance ($d_2 \approx 133.8\,\mathrm{mm}$) for a blurred dataset. Part 3 details the competition solution: by analyzing the angular distribution (θ_x) of the incident rays, we successfully disentangled three distinct objects superimposed in the hologram. The extracted images were identified as the WashU Crest, Department Chair Bruno Sinopoli, and a Boston Dynamics robot.**

I. Introduction

Optical imaging is traditionally described by geometrical optics, where light travels as rays. While complex optical systems can be difficult to model analytically, the paraxial approximation allows us to treat ray propagation as a linear system. This case study explores the implementation of such models in MATLAB to simulate cameras and holography [1].

The challenge lies in the nature of the provided "light field" dataset. Unlike a standard photograph, which captures only the intensity of light at a specific 2D plane, a light field captures the full 4D state of the rays: horizontal and vertical positions (x, y) and their propagation angles (θ_x, θ_y). The provided dataset contains 3×10^6 rays traveling at random angles from multiple sources. When these rays are projected directly onto a sensor without an optical system, they mix incoherently, resulting in a blurred, unrecognizable image.

To recover the information contained within this light field, we must "refocus" the rays computationally. By treating the rays as vectors and the optical elements as linear transformation matrices, we can mathematically simulate the bending of light by physical lenses [2].

The goals of this project are threefold: 1) To implement linear ray-transfer matrices for free space and lenses, verifying the physics of convergence and finite apertures. 2) To design an automated autofocusing system that recovers a sharp image from a blurred light field by optimizing the image plane distance. 3) To "crack" a holographic dataset by separating distinct objects based on their incident angles, mimicking the parallax effect observed in physical holograms (e.g., security stickers on credit cards) [1].

II. Theoretical Background

A. The Ray State Vector

In geometrical optics, a light ray at any point along the optical axis (z) is fully described by its position and angle relative to the axis. For a 3D system, we define the state vector $\mathbf{r}$ as:

$$\mathbf{r} = \begin{bmatrix} x & \theta_x & y & \theta_y \end{bmatrix}^T \tag{1}$$

where x, y are spatial coordinates and θ_x, θ_y are the angles of propagation in the xz and yz planes, respectively [2].

B. The Paraxial Approximation

To utilize linear algebra, we assume the paraxial approximation, where rays travel at small angles relative to the optical axis. This allows us to approximate $\tan(\theta) \approx \theta$ and $\sin(\theta) \approx \theta$ [2]. Under this assumption, the transformation of a ray by an optical element can be described by a linear matrix multiplication:

$$\mathbf{r}_{out} = \mathbf{M}\mathbf{r}_{in} \tag{2}$$

C. Ray-Transfer Matrices

We utilize two fundamental matrices to model the system [1]: **1. Free Space Propagation (M_d):** As a ray travels a distance d in free space, its position changes based on its angle ($x_{out} = x_{in} + d\theta_{in}$), while its angle remains constant.

$$M_d = \begin{bmatrix} 1 & d & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

2. Thin Lens (M_f): A thin lens changes the angle of a ray based on its position ($\theta_{out} = \theta_{in} - x_{in}/f$), while the position remains effectively unchanged as it passes through the thin element.

$$M_f = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1/f & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/f & 1 \end{bmatrix} \tag{4}$$

III. Part 1: Ray Tracing Validation

Before processing the light field data, we validated our matrix engine by simulating a known physical setup: a finite lens system.

A. System Setup

We simulated a thin lens with a focal length $f = 150\,\mathrm{mm}$ and a physical radius $r_{lens} = 20\,\mathrm{mm}$. Two point sources were placed at the object plane ($z = 0$): one on-axis at $(0, 0)$ and one off-axis at $(10\,\mathrm{mm}, 0)$. Each source emitted a fan of 8 rays with angles between $\pm\pi/20$ radians [1].

The rays propagated a distance $d_1 = 0.2\,\mathrm{m}$ to the lens. To form a focused image, the image distance d_2 must satisfy the thin lens equation:

$$\frac{1}{d_1} + \frac{1}{d_2} = \frac{1}{f} \implies d_2 = \left(\frac{1}{0.15} - \frac{1}{0.2}\right)^{-1} = 0.6\,\mathrm{m} \quad (5)$$

B. Implementation and Results

The total system matrix describing the ray path from object to image is given by the product of the component matrices, applied in reverse order of propagation:

$$\mathbf{r}_{image} = M_{d2} \cdot M_{finite} \cdot M_{d1} \cdot \mathbf{r}_{object} \quad (6)$$

A critical design choice in our simulation was the handling of the finite aperture. After the first propagation step (M_{d1}), we inspected the position x of every ray. We implemented a logical filter where any ray with $|x| > r_{lens}$ was assigned 'NaN'. This prevented "lost" rays from being refocused by the lens matrix, accurately simulating physical blocking [1].

Fig. 1 illustrates the results. The rays (blue and red) diverge in free space, are refracted by the lens, and converge to distinct points at the image plane $z = 0.8\,\mathrm{m}$ ($d_1 + d_2$). Rays striking outside the $20\,\mathrm{mm}$ radius (visible as the truncated fans) were correctly clipped.

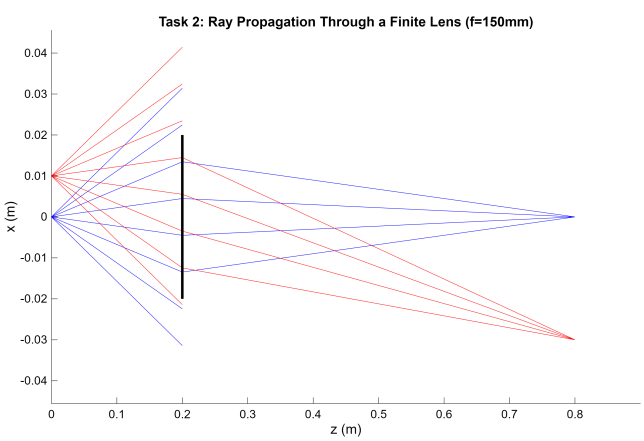


Fig. 1: Ray tracing diagram for Task 2. Rays originate at $z = 0$, pass through a lens at $z = 0.2$m, and converge at $z = 0.8$m. Note the outer rays terminating at the lens plane (black vertical line). This confirms that the matrix simulation correctly implements non-linear aperture constraints by clipping rays where $|x| > r_{lens}$.

IV. Part 2: Focusing the Light Field

A. The Blur Problem

Loading the raw 'lightField.mat' data and projecting it directly onto a sensor (using 'rays2img') resulted in a blurred, incoherent image. This blur occurs because rays originating from the same point on the object arrive at the sensor at different locations due to their differing propagation angles. In free space (M_d), there is no mechanism to redirect these diverging rays back to a single point. A lens (M_f) is required to map angle-to-position, effectively reversing the divergence [2].

B. Autofocus Algorithm Design

Since the distance to the object (d_1) was unknown, the required image distance (d_2) was also unknown. We designed an automated "focus hunting" algorithm similar to contrast-detection autofocus used in digital cameras.

1. Metric Definition: We defined our focus score as the standard deviation of the pixel intensities:

$$Score(d_2) = \mathrm{std}(\mathrm{image_pixels}) \quad (7)$$

This metric was chosen over other methods (such as gradient summation) because it is computationally efficient ($O(N)$) and robustly captures the high variance associated with sharp edges and distinct features in binary-style images.

2. Dynamic Sensor Scaling: A major challenge was that changing d_2 changes the magnification of the system ($M = -d_2/d_1$). A fixed sensor size would either crop the object or capture too much empty space, skewing the contrast metric. To solve this, we implemented a robust dynamic sensor width W:

$$W = 6 \times \max(\sigma_x, \sigma_y) \quad (8)$$

where σ_x, σ_y are the standard deviations of the ray positions at the current plane. This ensures the sensor always captures 6σ (99.7%) of the rays, maintaining a consistent framing.

3. Coarse-to-Fine Search: To ensure efficiency, we implemented a two-stage search:

- **Coarse Search:** Swept d_2 from $80\,\mathrm{mm}$ to $200\,\mathrm{mm}$ with $N = 200$ steps. This step size ($\Delta d \approx 0.6\,\mathrm{mm}$) was chosen to provide sub-millimeter resolution, ensuring the global peak was not skipped.
- **Fine Search:** Swept a narrow range $\pm 2\,\mathrm{mm}$ around the peak with high resolution.

C. Results

The algorithm converged on an optimal image distance of $d_2 \approx 133.8\,\mathrm{mm}$ (with $f = 100\,\mathrm{mm}$). However, even at this optimal focus, the resulting image appeared to be a chaotic superposition of multiple transparent objects. This indicated that the "blur" was not just a focus error, but a feature of the holographic data itself [1].

V. Part 3: Competition (Holographic Separation)

A. Theoretical Basis: The Parallax Effect

The superposition observed in Part 2 is characteristic of a hologram. Physical holograms record the light field such that the observed image changes based on the viewing angle. This is often seen in security holograms on credit cards, where tilting the card (changing the viewing angle) reveals different images [1].

In our ray-optics model, the "viewing angle" corresponds directly to the incident ray angle θ_x. We hypothesized that the dataset contained three distinct objects encoded into different angular sectors of the light field. By filtering the rays based on θ_x, we could computationally simulate "tilting" the hologram to isolate each object.

B. Algorithm Implementation

To separate the objects, we developed the following signal processing pipeline in 'CS3Separation.m':

1. Angular Histogram Analysis: We computed the histogram of the horizontal incident angles (θ_x) for all 3 million rays. As hypothesized, the distribution was not uniform. Fig. 2 reveals three distinct "clumps" of rays, separated by valleys of low density. These clusters correspond to the Left, Center, and Right viewing perspectives.

2. Automated Thresholding with Guard Bands: To separate these clusters, we identified the local minima between the peaks. A critical design choice was the implementation of **guard bands**. We did not simply cut the data at the midpoint. Instead, we defined a "dead zone" of $\Delta\theta = 20\,\mathrm{mrad}$ between the object ranges.

- *Rationale:* This value was empirically determined by observing the histogram valleys. A smaller gap included "ghosting" artifacts from adjacent views, while a larger gap significantly discarded valid signal rays. The $20\,\mathrm{mrad}$ buffer maximized the signal-to-noise ratio in the reconstructed images.

3. Reconstruction: The rays were split into three subsets based on the computed thresholds. Each subset was then independently propagated through the optimized lens system ($f = 100\,\mathrm{mm}, d_2 = 133.8\,\mathrm{mm}$) and rendered using 'rays2img'. We applied a final contrast stretch (clipping the bottom 1% and top 0.5% of intensities) to utilize the full dynamic range of the grayscale image.

C. Separation Results

The angular separation strategy was highly successful. Fig. 3 displays the three extracted components.

- **Object 1 (Left View):** Identified as the **Washington University in St. Louis Crest**. This object was encoded in the negative θ_x range.
- **Object 2 (Center View):** Identified as **Bruno Sinopoli**, the Chair of the Preston M. Green Department of Electrical & Systems Engineering. This object occupied the central angular region.

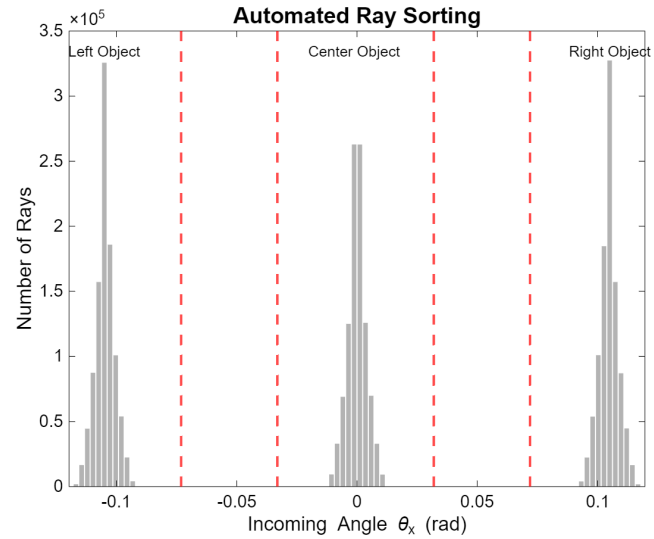


Fig. 2: Automated angle analysis histogram. The distribution of incident angles (θ_x) reveals three distinct clusters. The red dashed lines indicate the computed separation thresholds. The clear separation between modes validates the hypothesis that the hologram encodes spatial views into angular sectors.

- **Object 3 (Right View):** Identified as **Spot**, the robotic dog manufactured by Boston Dynamics. This object was encoded in the positive θ_x range.

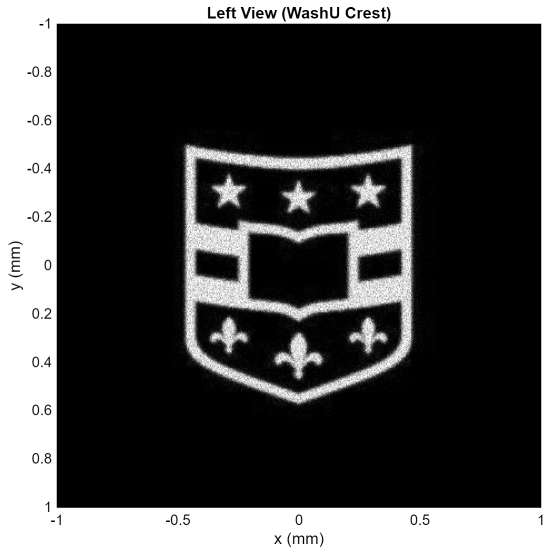
VI. Discussion and Limitations

The success of the angular separation confirms that the light field contained a "parallax-encoded" hologram. However, the simulation relies on the Forward Euler approximation and the assumption of ideal thin lenses. In a real physical system, lens aberrations (spherical, chromatic) would degrade the focus quality, particularly for the rays traveling at high angles (θ_x) which we relied upon for the Left and Right objects [2]. Furthermore, the resolution of the recovered images is fundamentally limited by the number of rays. While 3 million rays seems large, splitting them into three objects and spreading them over a 2D sensor introduces "shot noise" (graininess), which is visible in the recovered images (Fig. 3).

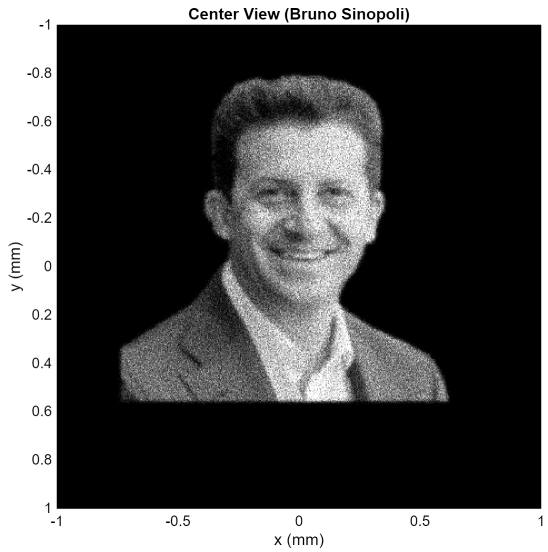
VII. Conclusion

This case study demonstrated the power of linear algebra in modeling optical systems. We successfully implemented a ray-tracing engine using matrix formalism, allowing us to simulate complex behaviors like focusing and finite apertures. By implementing an automated contrast-maximization loop, we solved the autofocus problem without prior knowledge of the object distance.

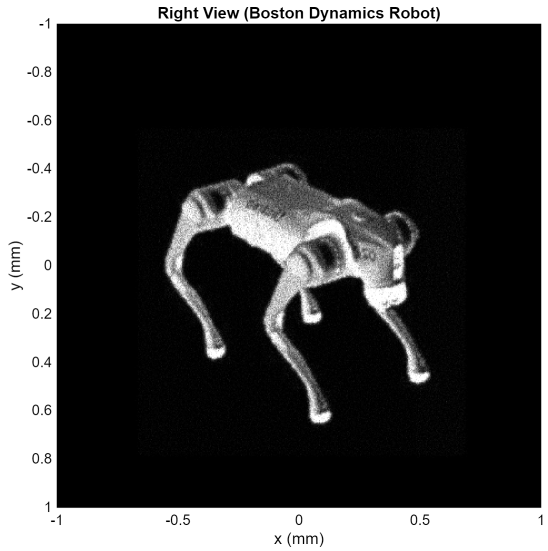
Most significantly, we solved the inverse problem of the holographic dataset. By leveraging the physical intuition that holograms encode spatial information into angular channels, we designed a filtering algorithm that successfully disentangled the superimposed light field. The clean separation of the WashU



(a) Left View ($\theta_x < -0.1$ rad): WashU Crest



(b) Center View ($\theta_x \approx 0$ rad): Bruno Sinopoli



(c) Right View ($\theta_x > 0.1$ rad): Boston Dynamics Spot

Fig. 3: The final extracted objects. By computationally isolating specific angular sectors of the light field, we successfully recovered the three distinct images hidden within the hologram.

Crest, Bruno Sinopoli, and the Spot robot validates both the theoretical model and the computational approach.

Member Contributions

Derek Nester:

- **Ray Tracing Engine & Vectorization:** Implemented the high-performance matrix propagation logic in 'CS3P1.m', including the handling of finite apertures and large-scale vector operations.
- **Autofocus Algorithm Design:** Developed the automated "focus hunting" logic, including the contrast maximization metric and the dynamic sensor width scaling system to handle magnification changes.
- **Holographic Separation:** Designed the signal processing pipeline for Part 3, including the angular histogram analysis and the "guard band" thresholding logic.
- **Report Writing:** Authored the technical sections: "Theoretical Background", "Autofocus Algorithm Design", and "Algorithm Implementation".

Naya Ford:

- **Simulation Validation:** Conducted the finite lens simulations in Part 1 to verify the basic physics of the model were functioning correctly.
- **Visualizations:** Generated all figures for the report, including the ray diagrams, the angular histograms, and the final high-contrast object images.
- **Report Writing:** Authored the introductory and concluding sections: "Introduction", "Discussion and Limitations", and "Conclusion".

References

[1] ESE 105 Case Study 3 Instructions, "Optical Imaging as a Linear System," Washington University in St. Louis, Fall 2025.

[2] B. E. Saleh and M. C. Teich, *Fundamentals of Photonics*, 3rd ed. Hoboken, NJ: John Wiley & Sons, 2019.